

White Shark Research in the Past: A Perspective

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There are over 350 species of sharks, with members of the group as diverse as the huge plankton-feeding whale and basking sharks, tiny deep-water sharks that are bioluminescent or that take "cookie cutter" bites out of seals and whales, and the bizarre hammerhead sharks, with their greatly expanded rostrum. Yet humans often ignore this great diversity and think of members of the group as conforming to a single image, that of the white shark *Carcharodon carcharias*. The public's fascination with this species is reflected by record attendance of the movie *Jaws* and its many sequels. The reason for this obsession is a morbid one. Humans consider the white shark dangerous—it feeds on us. One of us was possessed by the strong emotional response aroused by this predator 10 years ago while witnessing a predatory attack on a seal not more than 10 m from shore at Southeast Farallon Island. The water was crimson red with blood when the shark appeared, swimming back and forth, high in the water. The experience sent a shiver down the spine and cold sweat to the forehead. Although all of us regularly experience the human feelings of thirst, hunger, anger, and sexual arousal, rarely do we have the fear of being eaten. We feel this when watching a white shark devour a seal.

Despite the human obsession with this predator, we have been slower to learn about the biology of this species. Since World War II, many studies have been conducted on the biology of sharks, primarily members of the family Carcharhinidae. This has been due

to the interest and financial support of the U.S. Navy. However, it was only 10 years ago, on May 7, 1983, that a symposium was held at the California State University, Fullerton campus on the biology of the white shark. Although a few presentations at this meeting were quantitative, most sets of observations were small, and the other presentations consisted largely of anecdotal stories. Yet this meeting was a milestone. In any scientific discipline, we learn in a stepwise manner. Our first observations are anecdotal, only later to become increasingly quantitative as we gather sufficient information to begin to ask testable questions. Empirical studies then lead to experimental ones. After that first meeting, relatively little on the white shark appeared in the scientific literature until the symposium we organized at the Bodega Marine Laboratory of the University of California on March 4–7, 1993. This meeting drew scientists from six continents to talk on topics ranging from the evolution of the white shark to its behavior.

A favorite topic of the press, newspaper articles about white sharks often contain ideas rather than facts, and this has frustrated those who are willing to obtain data in the slow, methodical manner characteristic of science. It takes great restraint not to say something that seems correct but which is based on observations too few to make a firm conclusion. Disseminating such ideas can lead to trouble later; only in the process of disproving ideas does science advance closer to the truth.

The public's misconception regarding this mythical species even occurs in the name of the shark, "white shark." The first sharks that were seen were dead, lying on their back on the deck of a boat, and the species was given a popular name based on the coloration of its belly. However, from an ecological perspective, "black shark" is probably a better name. This black coloration matches the color of a rocky or vegetated bottom or the darkness over deep water. Adult sharks can swim under their favorite prey, seals and sea lions, without being detected and ambush them from below. Even the other popular name, "maneater," is called to question in this book. In Chapter 22, Klimley *et al.* hypothesize that sharks often spit out humans, birds, and sea otters after seizing them, because their bodies lack the energy-rich layers of fat possessed by pinnipeds and cetaceans.

In organizing the symposium on the biology of the white shark, we first wanted to provide an opportunity for scientists to make traditional scientific presentations. For this reason, we scheduled four sessions on different aspects of the biology of the shark. These resulted in papers that are now grouped to round out the eight sections of the book: Evolution, Anatomy, Physiology, Behavior, Ecology and Distribution, Population Biology, and Interactions with Humans.

However, we also wanted to provide a forum for evaluating several controversial ideas that had appeared repeatedly in newspapers and, for that reason, are believed by the public to be facts. One such

idea was that the population of white sharks off the western coast of North America was growing because seals and sea lions were steadily becoming more abundant. In contrast, sharks were decreasing off the eastern coast as they became increasingly overfished. In fact, this was happening to white sharks there as well as elsewhere in the world. There was concern that this could occur on the West Coast. Another common speculation was that white sharks attack people because they look like seals swimming at the surface of the water. This may seem obvious, but is not a tested fact yet—much disagreement exists in this book between the conclusions of Collier *et al.* (Chapter 19), Anderson *et al.* (Chapter 20), and Strong (Chapter 21). These and other controversial ideas were evaluated during the plenary sessions on population dynamics, predatory strategies, and attack avoidance. Participants gave brief talks, a moderator solicited questions from the audience, and a rapporteur gave a critical evaluation of the discussions presented. Three chapters in the book contain the comments of rapporteurs: Chapter 23 (by Barlow), Chapter 38 (by Cailliet), and Chapter 44 (by Baldrige). Through this process, we hoped to develop a consensus among knowledgeable persons on these very controversial topics.

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